Scalability & Future Plan

Date	05 October 2026
Team ID	SWTID-2026-2617
Project Name	EduGenie AI
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Gemini responses are sometimes slow	Some requests took 13 to 36 s; average 3.6 s against a 2 s target	High
2	Runs on a single local Uvicorn server	Requests wait in a queue when many users ask at once	High
3	No caching of repeated questions	Same question calls Gemini again, adding delay	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Tested with 5 users on a local server	Deploy on the cloud with multiple workers and a load balancer
2	Data Storage	No database used	Add a database for user history and quiz results
3	Performance	Avg 3.6 s, max 36 s	Async Gemini calls with timeout, and cache repeated answers
4	Security	API key kept in .env file	Use a secret manager, add user login and rate limiting

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact

Phase 2	Async Gemini calls with timeout and answer caching	Next 1 month	Faster and more stable response times
Phase 3	User login with progress and quiz history	2 to 3 months	Personalised learning for each learner
Phase 4	Cloud deployment with load balancing	4 to 6 months	Supports many more learners at once